PREPD – Dissertation Preparation

Mestrado em Engenharia Informática

Lecture 06

Ethics in research

Isabel Azevedo ifp@isep.ipp.pt

Dep. de Engenharia Informática – ISEP

2026/2027

Acknowledgement

Parts of this presentation are drawn from the resources cited in this document.

Overview

T1: Introduction

T2: Some events

T3: Deepening ethics in research

T4: Cases of ethical decision-making

T5: Ethics in your dissertation

This lecture covers ethics in research. Ethics in engineering is the subject of practical class 06, in the same week.

Before the class: play The Lab [15] as Kim Park, the graduate student (next two slides). It is discussed in T4.

Before the class: The Lab

The Lab: Avoiding Research Misconduct is an interactive movie produced by the Office of Research Integrity of the U.S. Department of Health and Human Services [15].

- You play the lead characters of a research laboratory and take decisions about research integrity – decisions with long-term consequences.
- It covers what this lecture is about: research misconduct, handling of data, responsible authorship, mentoring and questionable research practices.
- It makes the pressures visible: deadlines, hierarchy in the lab, competition for results and for authorship.

Why before the class? Because it is easier to discuss a decision that you have already had to take.

Before the class: The Lab – what to do

1. Open https://ori.hhs.gov/TheLab/ in a browser, with speakers or headphones (versions in English, Spanish, Chinese and Japanese).
2. Play as **Kim Park**, a third-year graduate student who questions the use of her data by another researcher.
3. Take your own decisions – there is no grade for choosing "well". If you want, go back and try another path.
4. Note down, to bring to the class:
   - the decision you found hardest, and why;
   - one situation that you consider research misconduct and one that you consider a questionable research practice;
   - what could have prevented the problem.

Discussed in T4 (Case 3).

T1: Introduction

T1: Ethics

- "Standards of conduct (or behaviour) that distinguish between right/wrong, good/bad, etc." [1].
- The study of standards of conduct.

T1: Applied ethics

- The study of ethics in specific situations, professions or institutions, e.g. medical ethics, research ethics, etc.

- In the PREPD course, there are two main topics:
  - research ethics (**this lecture**), and
  - engineering ethics (practical class 06).

From [1].

T1: Research ethics

"ethics underpins **all phases of research**, from when the idea for a project is conceived, all the way through its design and implementation, and on to how its findings are disseminated and put into practice in individual decisions or in policy" [16].

T2: Some events

T2: Notable events

Many events have shaped the way ethics in research is seen today, namely in these areas [19]:

- data integrity and research misconduct;
- authorship, credit and peer review;
- ethics of new and dual-use technology;
- human subjects protection.

T2: Data integrity and research misconduct

Fabrication, falsification or plagiarism [19]:

- **1830**: The mathematician Charles Babbage, often called the "father of the computer", wrote about dishonest data practices, including "fabricating, cooking, trimming, and fudging data".

- **1909**: The physicist Robert Millikan did not report 33 of the 149 observations of the oil-drop experiments for which he later received the Nobel Prize – those he had marked as "fair" or "poor".

T2: Data integrity and research misconduct (2)

Cases of data fabrication that led to retracted papers [19]:

- **1974**: William Summerlin admitted to fabricating data by using a marker pen to make black patches on white mice.
- **2002**: Jan Hendrik Schön, a rising star in physics, was found to have fabricated and falsified data in dozens of papers. Many of his papers were retracted.
- **2014**: Haruko Obokata was found to have fabricated and falsified data in high-profile stem-cell papers, which were retracted.

Other stem-cell papers under scrutiny, in 2008 and in 2024:

- 'Manipulated' stem-cell paper faces retraction (2008);
- Retraction note: Pluripotency of mesenchymal stem cells derived from adult marrow (2024).

T2: Data integrity and research misconduct (3)

- **Reproducibility** [19]:
  - **1989**: The "cold fusion" announcement by Pons and Fleischmann collapsed when dozens of laboratories could not reproduce their results.
  - **2013**: The NIH launched an initiative to promote rigour, transparency and reproducibility in research.
- **Warnings that were ignored** [19]:
  - **1986**: Roger Boisjoly, an engineer at Morton Thiokol, warned that the O-ring seals of the Space Shuttle Challenger could fail if it was launched in cold weather. His warnings were overruled, and the shuttle broke apart shortly after lift-off, killing its seven crew members [17].

Authorship and credit [19]:

- **1909**: Robert Millikan did not name his student, Harvey Fletcher, as an author, despite Fletcher's important contributions to the oil-drop experiments.
- **1953**: Rosalind Franklin's X-ray diffraction image of DNA ("Photo 51") was crucial for verifying the model proposed by James Watson and Francis Crick. It was shown to them without her permission.
- **1997**: The ICMJE revised its guidelines: to be an author, one must make a substantial contribution to the research design or to data collection, analysis or interpretation, and write the paper or revise it critically.

Failures of peer review and of editorial control [19]:

- Hoaxes **(1996)**: The physicist Alan Sokal submitted a paper filled with nonsense to a cultural studies journal to test its rigour. It was accepted and published.
- Predatory journals **(2010)**: Jeffrey Beall published a list of "predatory journals", which charge high publication fees but provide poor or non-existent peer review.
- Rushed publication **(2020)**: During the COVID-19 pandemic, the rush to publish led to the dissemination of invalid results and to multiple retractions, such as those of the papers based on data from the company Surgisphere.

T2: Ethics of new and dual-use technology

- Scientist-led governance **(1973–1975)**: Scientists met to discuss the risks of the new recombinant DNA technology and to establish biosafety protocols.
- Dual-use research **(2014)**: Research whose results can be used both for good and to cause harm. The concerns over bioterrorism and gain-of-function research highlight the dilemma of publishing information that could be misused.
- Generative AI **(2024)**: The use of generative AI in research has raised new ethical issues concerning accountability, rigour, reproducibility, honesty and the allocation of credit.

Adapted from [19].

T2: Human subjects protection

- Nuremberg Code **(1947)**: The first international code of research ethics. It states that "the **voluntary consent** of the human subject is absolutely essential" [13].
- Oversight **(1974)**: In the USA, the National Research Act, passed after hearings on the Tuskegee study and signed in 1974, authorised federal agencies to develop regulations for research with human subjects.
  - These require institutions to form **Institutional Review Boards (IRBs)** to review and oversee research with human subjects.
- The Common Rule **(1991)**: The basic regulatory framework adopted by US federal agencies for the protection of human subjects in research.

Adapted from [19].

T3: Deepening ethics in research

T3: Ethics = law?

Most societies also have legal rules that govern behaviour, but **ethical norms tend to be broader and more informal than laws**. Although most societies use laws to enforce widely accepted moral standards, and ethical and legal rules use similar concepts, ethics and law are not the same. An action may be legal but unethical, or illegal but ethical.

From [18].

Ethics Versus the Law: The Case of the Belfast Project

- The Belfast Project, at Boston College, recorded testimonies from several sides of the conflict in Northern Ireland in the late 20th century. The participants received a **written guarantee that their testimonies would be kept secret until after their deaths** – a guarantee that could not be upheld when some of the recordings were subpoenaed.
- "there is even uncertainty about whether research institutions will keep their promises when facing external pressure"
- "it is not possible to fully inform someone about the risks they would be taking if they participate in research"

From [11].

More details in https://www.bbc.com/news/uk-northern-ireland-27238797.

T3: Ethics in project/thesis/dissertation/internship work

The Code of Good Practices and Conduct of P.PORTO (Despacho 11171/2020) [8] addresses some aspects in specific articles, namely:

- Article 6: Duties of students (point *n*);
- Article 8: Declaration of commitment;
- Article 10: Good practices in research activities.

T3: Ethics in project/thesis/dissertation/internship work (2)

"As pessoas que desenvolvam investigação devem ainda:

- Assegurar os mais elevados padrões éticos e legais em todas as atividades de investigação, nomeadamente quando inclua a **participação de seres humanos ou animais, a colheita e processamento de dados pessoais sensíveis, a utilização de células/tecidos de origem humana, os sistemas suscetíveis de operação autónoma, em particular quando envolva o uso de inteligência artificial, ou comunidades/pessoas desfavorecidas**;

- Submeter as atividades de investigação, referidas na alínea anterior, a apreciação por comissão de ética com competências na área específica, conforme imposição legal;"

From [8].

T3: Ethics in project/thesis/dissertation/internship work (3)

- From the previous slide: "Submeter as atividades de investigação, referidas na alínea anterior, a **apreciação por comissão de ética com competências na área específica**, conforme imposição legal"

- Some work may therefore require approval from an ethics review body. Example: the Ethics Committee of the School of Health of P.PORTO, for clinical research.

From [8].

T3: Ethics in project/thesis/dissertation/internship work (4)

Transparency and reproducibility – two examples of work whose artefacts are publicly available:

- Final-year undergraduate project
- Master's dissertation project

T3: Research integrity

- Deal fairly with others.
- Be honest about the methods and the results.
- Allow the results to be replicated, wherever possible.
- Protect the welfare of research subjects.
- ...

From [4].

T3: The bias problem in scientific research

- Bias in scientific research refers to systematic errors or deviations that occur due to the influence of factors other than the phenomenon under investigation.
- To minimise bias, it is essential to use rigorous research methods and to be transparent about all aspects of the research process.

From [10].

T3: Common types of bias

- **Sampling bias**: Is the sample representative of the population being studied? Research may be based on a limited or unrepresentative set of software systems, developers or environments, which can skew the findings. For example, if studies focus mostly on open-source projects, the results may not generalise well to closed-source enterprise environments.

- **Confirmation bias**: Researchers may unconsciously favour results that confirm their hypotheses, and overlook data that contradict their assumptions.

Based on [10].

T3: Types of research misconduct

- **Fabrication**: making up data or results.
- **Falsification**: manipulating research materials, equipment or processes, or changing or omitting data, so that the research is not accurately represented.
- **Plagiarism**: using the ideas, results or words of others without giving appropriate credit.

Honest error and differences of opinion are not research misconduct [18].

Based on [4, 1].

T3: Fabrication vs falsification

- **Fabrication**: "making up data or results and recording or reporting them" – **nothing real** lies behind the data.
- **Falsification**: "manipulating research materials, equipment, or processes, or changing or omitting data or results" so that the research is not accurately represented – the research **was real, but was altered**.

The question to ask: was there anything real behind the data?

- Reporting results of tests that were never run – **fabrication**.
- Altered images of real experiments (Kim Park's case in The Lab) – **falsification**.
- Millikan: omitting observations would be **falsification**, if intentional and misleading.
- Schön, Obokata: both [19].
- Summerlin painted the mice to show skin grafts that had not succeeded: **fabrication or falsification?**

Definitions from [21].

T3: Ethical concerns in research

Ethical concerns relate to the principles and standards that ensure that research is conducted responsibly, with integrity, and in ways that respect the rights and well-being of all those involved.

Based on [18].

T3: Ethical concerns in research (2)

- **Informed consent**: ensuring that participants, if any, are fully aware of the purpose of the research and agree voluntarily to take part.
- **Privacy and data security**: ensuring that sensitive information (e.g. user data, system logs) is handled securely, particularly when real-world datasets are used.
- **Bias and fairness**: ensuring that the algorithms or systems being researched do not discriminate against any group or individual.
- **Misuse of technology**: avoiding the development of technologies that could cause harm, be misused or lead to unintended consequences.
- **Transparency**: presenting all the results (positive, negative or neutral) to maintain academic integrity.

Based on [18, 20, 7].

T3: Other unacceptable practices

- Hiding the use of AI or automated tools in the creation of content or drafting of publications.
- Citing selectively or inaccurately.

From [20].

T3: European Code of Conduct for Research Integrity

The European Code of Conduct describes **professional, legal, societal, ethical, and moral responsibilities** of the different actors in different settings, including those who define and implement the priorities and criteria for research funding, assessment, and publication.

From [20].

T3: Surveys

The results obtained through a survey can be compromised if there is a possibility of:

- bias: for instance, co-workers commenting on the work of one of them;
- lack of confidentiality: respondents who can be identified may be influenced in their answers.

From [4].

T3: Ethics evaluation guide

- Do the researchers discuss the ethics of their research and whether they were guided by any code of ethical conduct? If not, how does that affect your confidence in the research?
- Do the researchers explain any ethical dilemmas they faced and how they resolved them?
- Are there additional ethical issues that the researchers would have needed to address?

From [14].

T3: Ethics evaluation guide (2)

Do the researchers state the remaining uncertainties of their results, and the problems that might occur if they are applied to products and services? Suggestion: Summarise the outcomes in a concluding section, but state the uncertainties and problems which might occur if the results are applied to products and services.

- European Code of Conduct for Research Integrity [20]:
  - section 2.7: authors are transparent about "the robustness of the evidence, including remaining uncertainties and knowledge gaps";
  - section 2.4: "Researchers recognise and weigh potential harms and risks relating to their research and its applications".
- ACM Code of Ethics and Professional Conduct, principle 2.5 [3]: "Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks."

T4: Cases of ethical decision-making

T4: Case 1

The research protocol for a study of a hypertension drug requires the administration of the drug at different doses to 50 laboratory mice, with chemical and behavioural tests to determine toxic effects.

Tom has almost finished the experiment for Dr Q. He has only 5 mice left to test. However, he really wants to finish his work in time to go to Florida on spring break with his friends, who are leaving tonight. He has injected the drug in all 50 mice, but has not completed all of the tests. He therefore decides to extrapolate from the 45 completed results to **produce the 5 additional results**.

Fabrication?

From [18].

T4: Case 2

Dr T has just discovered a **mathematical error** in his paper that has been accepted for publication in a journal. The error does not affect the overall results of his research, but it is potentially misleading. The journal has just gone to press, so it is too late to catch the error before it appears in print. In order to avoid embarrassment, Dr T decides to **ignore the error**.

Misconduct?

From [18].

T4: Case 3 – The Lab

The Lab: Avoiding Research Misconduct is an interactive movie in which the viewer takes decisions, with long-term consequences, about issues of ethical conduct of research from a variety of perspectives: a graduate student, a postdoctoral researcher, a principal investigator and a research integrity officer [15].

Played before the class as Kim Park, the graduate student (https://ori.hhs.gov/TheLab/):

- Which decision was the hardest? Why?
- Which situations were research misconduct (fabrication, falsification, plagiarism), and which were questionable research practices?
- What would have prevented the problem: the mentor, the data records, the authorship rules?

Misconduct?

T5: Ethics in your dissertation

T5: Generative AI in the dissertation

Recommendations for researchers [5]:

- **You remain ultimately responsible** for the scientific output. AI systems are neither authors nor co-authors.
- **Use generative AI transparently**: declare the tools used substantially – which tool, which version, for what and how it affected the work – usually in the methods chapter.
- **Check what it produces**: generative AI may produce plausible-sounding but incorrect citations and summaries, and its answers reflect the biases of its training data.
- **Privacy, confidentiality and intellectual property**: do not upload unpublished work, personal data or the data and code of a company to external AI tools without guarantees that they will not be reused.

Hiding the use of AI or automated tools is an unacceptable practice [20].

T5: Generative AI – what to agree with your supervisors

To discuss and agree with the supervisors, task by task, before using it:

- help with the literature review (searching, summarising, comparing papers)? The papers must still be read;
- help with writing, translation or revision of the text?
- generation or refactoring of code, tests or scripts? Who verifies it, and under which licence?
- generation of results of experiments that were not carried out? Presented as results, this is **fabrication** [1]; synthetic data or simulations are acceptable only when agreed, clearly identified and described in the methods.

What should not even need discussing: fictitious bibliographic references. In one study, 55% of the citations generated by GPT-3.5 and 18% of those generated by GPT-4 were fabricated [22]. Every reference must exist and must have been read.

T5: Plagiarism in practice

Plagiarism is "using other people's work or ideas without giving proper credit to the original source" [20] – not only copying a paragraph. It also includes:

- using ideas, sentences or texts of others without citing them [8] – also when they are paraphrased or translated;
- reusing figures, tables or code without attribution [1] (or against their licence);
- reusing one's own work without citation ("self"-plagiarism), which "may be unethical and may violate copyright laws" [1]; the P.PORTO code forbids presenting as original a work already presented or published, without saying so [8].

At ISEP, the dissertation must be **original** and the student declares that the text is of their **exclusive authorship**, with the contributions of others referenced; failing to comply is a disciplinary offence [9].

T5: Personal data in the dissertation

Surveys, interviews, usage logs or user data are **personal data** [6, 2]:

- have a lawful basis – in research with participants, usually their **informed consent** – and tell them what is collected, why, for how long and who has access;
- collect only what is needed (data minimisation) and define for how long it is kept;
- pseudonymised data are **still personal data**; only anonymous data fall outside the GDPR;
- store the data securely, and never in personal accounts or external AI tools.

Sensitive data or vulnerable participants: check whether an ethics committee must assess the work (T3).

T5: Dissertations in a company

- The work is supported by a **collaboration protocol** between the company and ISEP [9].
- **Confidentiality:** confidentiality, for up to three years, can be requested when the examinations are requested; it must be validated and justified by the jury [9].
- **Intellectual property:** who owns the code and the results? Agree it in writing at the start – the regulation does not settle it [9].
- **What can be published:** agree early what may be described in the dissertation and in papers, and what must stay out.
- **Conflict of interest:** a personal or financial interest "likely to bias" judgement [1] – for example, evaluating a product that you develop for the company that pays you. Declare it.

T5: Research data

Data and code should be **FAIR** [23]:

- **F**indable: a persistent identifier (e.g. a DOI) and metadata;
- **A**ccessible: retrievable through a standard protocol, even when access is restricted;
- **I**nteroperable: open, documented formats;
- **R**eusable: a licence, the provenance and enough documentation to reuse them.

Decide at the start where the data and code are kept and versioned, what can be published (and under which licence), and what cannot – because of personal data or confidentiality. FAIR does not mean open: it means well managed [12].

Closing

Three things to carry forward:

- ethics underpins **all the phases** of research – from the idea to the dissemination of the results – and not only the moment of data collection;
- **legal and ethical are different questions**: an action may be legal but unethical, or illegal but ethical;
- in the dissertation: no fabrication, falsification or plagiarism – nor fictitious references; the use of generative AI agreed with the supervisors and declared; and, when the work involves people, personal data or a company, the rules agreed before the work starts.

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